

Bihar Engineering University, Patna

B.Tech 1st Semester Exam-2022

Subject: Chemistry

Course: BTech.

Time: 03 Hours

Code: 100103

Full Marks: 70

Instructions:

- (i) The marks are indicated in the right-hand margin.
- (i) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.
- (v) Symbols used (if any) have their usual meanings.

Q. 1 Answer any seven of the following:

- (a) What is the designation of the orbital having $n = 4$ and $l = 3$?
- (b) Write the ground state electronic configuration of N_2^-
- (c) Which of Cr^+ or Cu^+ is expected to be coloured?
- (d) Arrange molecular species N_2 , N_2^+ , N_2^- and N_2^{2-} in increasing order of stability.
- (e) Out of Cr^{2+} and Cr^{3+} which one is stable in aqueous solution?
- (f) What is the direction of a reaction when $\Delta G = 0$?
- (g) A gas expands against vacuum. What is the work done on it?
- (h) What is the condition for a reaction to be in equilibrium?
- (j) Which of the following is not a nucleophile?

H_2O , BE_3 , NH_3 , OH^-

- (i) A reaction has $\Delta H < 0$ and $\Delta S < 0$. At what temperature the forward reaction proceed?

Q.2 At what temperature will water boil when the applied pressure is 528 mm of Hg? (Latent heat of vaporisation of water = 545.5 cal/g)

(b) Explain dual nature of light and give one example (property/experiment) in favour of its particle nature and wave nature.

(c) The equilibrium constants for the reaction $\text{H}_2(\text{g}) + \text{S}(\text{s}) \rightleftharpoons \text{H}_2\text{S}(\text{g})$ are 18.5 at 925K and 9.25 at 1000 K. Calculate standard enthalpy of the reaction. Also calculate ΔG° and ΔS° at 925 K.

Q.3 Which of the following two molecules has a higher bond length?

(i) O_2

(ii) O_2^+

(iii) O_2^-

Explain using molecular orbital theory.

(b) Draw the MO energy level diagram for NO molecule. Using this diagram, calculate and explain bond order and magnetic behaviour of (i) NO, (ii) NO^+ and (iii) NO^- .

Q. 4 (a) Calculate the frequency (in Hz and cm^{-1}) of O - H bond, if the force constant and reduced mass of the atom pair are 770 N m^{-1} and $1.563 \times 10^{-27} \text{ kg}$ respectively.

(b) Microwave spectrum of gaseous HCl molecule exhibits a series of equally spaced lines with interspacing of 20.7 cm^{-1} . Calculate the inter-nuclear distance of HCl molecule.

(c) Explain geometrical isomerism and optical isomerism for transition metal complex with an example for each.

Q.5 (a) Calculate the force constant of CO molecule, if its fundamental vibrational frequency is 2140 cm^{-1} . (At. Mass of carbon = $1.99 \times 10^{-26} \text{ kg}$ and O = $2.66 \times 10^{-26} \text{ kg}$.)

(a) At what frequency shift from TMS, would a group of nuclei with $\delta = 1.00$ resonate in an NMR spectrometer operating at 500 MHz?

(c) How many ^1H NMR signals are there in the following?

(i) $\text{CH}_3 - \text{CH}_3$

(ii) $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

(iii) $\text{CH}_3 - \text{CH}_2 - \text{Cl}$

(iv) $\text{CH}_3 - \text{CHCl} - \text{CH}_3$

(v) $\text{C}_6\text{H}_5\text{CH}_3$

(vi) $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3$

Q. 6 (a) 2 mol of NH_3 at 300K occupy a volume of $5 \times 10^{-3} \text{ m}^3$. Calculate the pressure using van der Waals equation ($a = 0.417 \text{ Nm}^4 \text{ mol}^{-2}$ and $b = 0.037 \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$). Compare the above result with the pressure calculated using ideal gas equation.

(b) Write short notes on the following:

(i) Magnetic resonance imaging

(ii) Fingerprint region in infrared spectroscopy

(iii) Different types of electronic excitations

Q. 7 (a) Write the principle for lime soda process for softening of hard water.

(b) Calculate the amount of lime and soda required for the softening of a million litres of hard water containing $\text{CaCO}_3 = 25 \text{ ppm}$, $\text{MgCO}_3 = 144 \text{ ppm}$, $\text{CaCl}_2 = 111 \text{ ppm}$, $\text{MgCl}_2 = 95 \text{ ppm}$, $\text{Na}_2\text{SO}_4 = 15 \text{ ppm}$, $\text{Fe}_2\text{O}_3 = 25 \text{ ppm}$.

(c) The hardness of 50000 litres of water sample was removed by passing it through a zeolite softener. The softener then required 200 L of NaCl solution, containing 125g/L of NaCl for regeneration. Calculate the hardness of the sample of water.

Q.8 (a) Write notes on the following:

(i) Optical isomerism of lactic acid

(ii) Optical isomerism of tartaric acid

(b) Differentiate between the following:

(i) Enantiomers and diastereomers

(ii) Racemic mixture and meso-compounds

Q.1 Choose the correct answer of the following: (any)

(a) What does the following declaration mean? `Int *ptr [5]`

(i) ptr is an array of pointers to 5 integers

(ii) ptr is an array of 5 integers

(iii) ptr is a pointer to an array of 5 integers

(iv) ptr is a pointer to array

(b) The function `malloc()` is declared in which header file.

Bihar Engineering University, Patna
B.Tech. 1st Semester Examination, 2023

Course: B.Tech.

Code: 100103

Subject: Chemistry

Time: 03 Hours

Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Answer the following (Any seven question only):

- (a) Why are Br^+ and CCl_3 electrophiles? [2 x 7 = 14]
- (b) What do you mean by single electrode potential?
- (c) Arrange molecular species N_2 , N_2^+ and N_2^- in increasing order of stability.
- (d) A water sample contains 204 mg of CaSO_4 per litre. Calculate the hardness in terms of CaCO_3 equivalent?
- (e) What is the value of effective nuclear charge (z^*) for last electron of Cu +? (Atomic no. of Cu=29)
- (f) Out of Cr^{2+} and Cr^{3+} , which one is stable in aqueous solution?
- (g) What is the direction of a reaction, when $\Delta G = 0$
- (h) How is the bond order related with dissociation energy?
- (i) The specific conductance of a decinormal solution of KCl at 18 °C is $0.0112 \text{ ohm}^{-1}\text{cm}^{-1}$. The resistance of the cell containing the solution at 18 °C was found to be 55 ohms. What is the cell constant?
- (j) A gas expands against vacuum. What is the work done on it?

Q.2 (a) Discuss Heisenberg's uncertainty principle and deduce a mathematical expression involving position and velocity of uncertainty of an electron. [7]

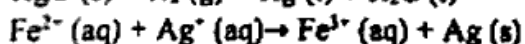
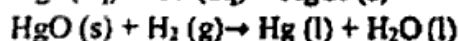
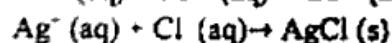
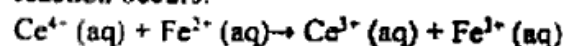
(b) Explain dual nature of light and give one example (property/experiment) in favour of its particle nature and wave nature. [7]

Q.3 (a) Draw the MO energy-level diagram for O_2 and based on the diagram explain the paramagnetic property observed in O_2 . [7]

(b) Explain ionization isomerism and linkage isomerism for transition metal complex with an example of each. [7]

Q.4 (a) 1 mole of an ideal gas expands isothermally and reversibly from 5 dm^3 to 10 dm^3 at 300 K. Calculate q , w , ΔU , ΔH , ΔG and ΔA . [7]

(b) Design electrochemical cells (in proper notation) in which each of the following reaction occurs: [7]



- Q.5 (a) Write the principle for lime soda process for softening of hard water? [7]
 (b) A water sample has the following constituents per litre. $\text{CaCO}_3 = 81 \text{ mg}$, $\text{Mg}(\text{HCO}_3)_2 = 75 \text{ mg}$, $\text{CaSO}_4 = 136 \text{ mg}$, $\text{MgSO}_4 = 120 \text{ mg}$, $\text{NaCl} = 4.7 \text{ mg}$. [7]
 Calculate the quantity of temporary and permanent hardness in the water sample.
 Calculate the quantity of lime (78% purity) and soda (92% purity) required for softening of 1.5 million litres of the water sample.
- Q.6 (a) The internuclear distance of NaCl is $2.36 \times 10^{-10} \text{ m}$. Calculate the reduced mass and moment of inertia of NaCl. (Atomic mass of $\text{Cl} = 35 \times 10^{-3} \text{ kg mol}^{-1}$ and $\text{Na} = 23 \times 10^{-3} \text{ kg mol}^{-1}$) [7]
 (b) Write the product for following reactions together with reaction mechanism: [7]
 (i) $2\text{C}_6\text{H}_5\text{COCH}_3 + \text{OH}^- \rightarrow$
 (ii) $\text{CH}_3\text{CH}_2\text{OH}$ (heated with H_2SO_4) $\rightarrow$
- Q.7 (a) Discuss photoelectric effect and explain equations involved with this effect. [7]
 (b) How many photons of light having a wavelength of $4000 \text{ \AA}$ are necessary to provide 1 J of energy? ($h = 6.626 \times 10^{-34} \text{ J-s}$) [7]
- Q.8 Explain the following: [3½ x 4 = 14]
 (a) $\text{S}_\text{N}1$ leads to racemic mixture, whereas $\text{S}_\text{N}2$ gives rise to inverted product.
 (b) o-nitrophenol easily separates from p-nitrophenol by steam distillation.
 (c) The amino group in aniline is o- and p-directing, but amide group is m-directing.
 (d) Benzoic acid is weaker than formic acid.
- Q.9 (a) Write notes on the following: [3½ x 2 = 7]
 (i) Optical isomerism of lactic acid
 (ii) Diels – Alder reaction.
 (b) Differentiate between the following: [3½ x 2 = 7]
 (i) Enantiomers and diastereomers
 (ii) Racemic mixture and meso-compounds

<https://www.akubihar.com>

Whatsapp @ 9300930012

Send your old paper & get 10/-

अपने पुराने पेपर्स भेजे और 10 रुपये पायें,

Paytm or Google Pay से

<https://www.akubihar.com>